

# Comprehensive Notes: Systems & Matrices

Echelon Form • Rank of Matrix • Conditions for Consistency • Gauss Elimination Method

## 1. Row Echelon Form (REF) & RREF

### Definition of Row Echelon Form (REF)

A matrix is in **Row Echelon Form (REF)** if it satisfies three structural conditions:

- **Zero Rows at Bottom:** All rows consisting entirely of zeros (if any) are located below all non-zero rows.
- **Leading Entry Position:** The first non-zero entry (called the *leading entry* or *pivot*) in any non-zero row is strictly to the right of the leading entry of the row above it.
- **Zeros Below Pivots:** All entries in a column directly below a leading pivot are zero.

**Reduced Row Echelon Form (RREF):** Further requires that every leading pivot equals 1, and each leading pivot is the *only* non-zero entry in its column.

### Standard Elementary Row Operations (EROs)

We perform EROs on a matrix to reduce it to Echelon form without altering its solution set:

1. **Row Interchange:**  $R_i \leftrightarrow R_j$  — Swap row  $i$  and row  $j$ .
2. **Non-Zero Scalar Multiplication:**  $R_i \rightarrow k \cdot R_i$  (where  $k \neq 0$ ).
3. **Row Addition / Subtraction:**  $R_i \rightarrow R_i + k \cdot R_j$  — Add a multiple of row  $j$  to row  $i$ .

## 2. Rank of a Matrix

The **Rank of a Matrix A** (denoted as  $\rho(A)$  or  $\text{rank}(A)$ ) is defined as:

- The **number of non-zero rows** in its Row Echelon Form.
- Equivalently, the maximum number of *linearly independent* row vectors or column vectors in A.

### Fundamental Properties of Rank:

- For an  $m \times n$  matrix A:  $0 \leq \text{rank}(A) \leq \min(m, n)$ .
- $\text{rank}(A) = 0$  if and only if A is a Zero Matrix.
- $\text{rank}(A) = \text{rank}(A^T)$ .
- Elementary row operations preserve the rank of a matrix.

### Example: Finding Matrix Rank via REF Reduction

Find the rank of matrix  $A$ :

$$A = \begin{bmatrix} 1 & -1 & 2 & 3 \\ 2 & 2 & 0 & 2 \\ 4 & 0 & 4 & 8 \end{bmatrix}$$

**Step 1:** Apply  $R_2 \rightarrow R_2 - 2R_1$  and  $R_3 \rightarrow R_3 - 4R_1$

$$\begin{bmatrix} 1 & -1 & 2 & 3 \\ 0 & 4 & -4 & -4 \\ 0 & 4 & -4 & -4 \end{bmatrix}$$

**Step 2:** Apply  $R_3 \rightarrow R_3 - R_2$  and scale  $R_2 \rightarrow \frac{1}{4} R_2$

$$\begin{bmatrix} 1 & -1 & 2 & 3 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

**Result:** There are **2 non-zero rows** in Echelon Form. Hence,  $\text{rank}(A) = 2$ .

## 3. Systems of Linear Equations in Three Variables

A system of three linear equations in three unknown variables ( $x, y, z$ ) takes the standard form:

$$a_1x + b_1y + c_1z = d_1$$

$$a_2x + b_2y + c_2z = d_2$$

$$a_3x + b_3y + c_3z = d_3$$

In matrix notation:  $AX = B$ , where:

$$A = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix} \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \quad B = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}$$

The **Augmented Matrix** combining coefficient matrix  $A$  and constant vector  $B$  is written as  $[A \mid B]$ .

## 4. Conditions for Consistency & Homogeneity

Let  $n = 3$  be the number of unknown variables  $(x, y, z)$ .

### A. Non-Homogeneous System ( $AX = B$ , where $B \neq 0$ )

*Rouché–Capelli Theorem Rules:*

| Condition                   | Rank Relationship | System Classification    | Nature of Solution                 |
|-----------------------------|-------------------|--------------------------|------------------------------------|
| $rank(A) = rank([A B]) = n$ | $r = r' = 3$      | Consistent & Independent | Exactly ONE Unique Solution        |
| $rank(A) = rank([A B]) < n$ | $r = r' < 3$      | Consistent & Dependent   | INFINITELY MANY Solutions          |
| $rank(A) \neq rank([A B])$  | $r \neq r'$       | Inconsistent             | NO Solution (Contradictory system) |

### B. Homogeneous System ( $AX = 0$ )

Since the zero vector is always in the column space,  $rank(A) = rank([A|0])$  always holds. Thus, a homogeneous system is **always consistent**.

| Determinant                 | Rank Condition | Solution Type                                             |
|-----------------------------|----------------|-----------------------------------------------------------|
| $ A  \neq 0$ (Non-singular) | $rank(A) = 3$  | Trivial Solution Only: $(x, y, z) = (0, 0, 0)$            |
| $ A  = 0$ (Singular)        | $rank(A) < 3$  | Non-Trivial Solutions: Infinitely many non-zero solutions |



### Example 1: Non-Homogeneous — Consistent with UNIQUE Solution

Solve the following system using Gauss Elimination:

$$x + y + z = 6$$

$$2x - y + z = 3$$

$$x + 2y + 3z = 14$$

#### Step 1: Write Augmented Matrix $[A|B]$

$$\left[ \begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 2 & -1 & 1 & 3 \\ 1 & 2 & 3 & 14 \end{array} \right]$$

#### Step 2: Reduce to Echelon Form

Apply  $R_2 \rightarrow R_2 - 2R_1$  and  $R_3 \rightarrow R_3 - R_1$ :

$$\left[ \begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & -3 & -1 & -9 \\ 0 & 1 & 2 & 8 \end{array} \right]$$

Swap  $R_2 \leftrightarrow R_3$  for a cleaner pivot:

$$\left[ \begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 8 \\ 0 & -3 & -1 & -9 \end{array} \right]$$

Apply  $R_3 \rightarrow R_3 + 3R_2$ :

$$\left[ \begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 8 \\ 0 & 0 & 5 & 15 \end{array} \right]$$

#### Step 3: Check Rank & Consistency

$\text{rank}(A) = 3$ ,  $\text{rank}([A|B]) = 3$ . Since  $\text{rank}(A) = \text{rank}([A|B]) = n = 3$ , the system has a **Unique Solution**.

#### Step 4: Back Substitution

- From Row 3:  $5z = 15 \Rightarrow z = 3$
- From Row 2:  $y + 2(3) = 8 \Rightarrow y + 6 = 8 \Rightarrow y = 2$
- From Row 1:  $x + 2 + 3 = 6 \Rightarrow x + 5 = 6 \Rightarrow x = 1$

**Final Solution:**  $(x, y, z) = (1, 2, 3)$

### Example 2: Non-Homogeneous — Consistent with INFINITELY MANY Solutions

Solve the system:

$$\begin{aligned}x + 2y - z &= 3 \\2x + 4y + z &= 9 \\3x + 6y &= 12\end{aligned}$$

#### Step 1: Form Augmented Matrix $[A|B]$

$$\left| \begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 2 & 4 & 1 & 9 \\ 3 & 6 & 0 & 12 \end{array} \right|$$

#### Step 2: Reduce to Echelon Form

Apply  $R_2 \rightarrow R_2 - 2R_1$  and  $R_3 \rightarrow R_3 - 3R_1$ :

$$\left| \begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 0 & 0 & 3 & 3 \\ 0 & 0 & 3 & 3 \end{array} \right|$$

Apply  $R_3 \rightarrow R_3 - R_2$ :

$$\left| \begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 0 & 0 & 3 & 3 \\ 0 & 0 & 0 & 0 \end{array} \right|$$

#### Step 3: Analyze Rank

$\text{rank}(A) = 2$ ,  $\text{rank}([A|B]) = 2$ . Since  $\text{rank}(A) = \text{rank}([A|B]) = 2 < 3$ , the system is **consistent with infinitely many solutions**.

#### Step 4: Back Substitution & Free Parameter

- From Row 2:  $3z = 3 \Rightarrow z = 1$
- From Row 1:  $x + 2y - 1 = 3 \Rightarrow x + 2y = 4$
- Let  $y = k$  (where  $k \in \mathbb{R}$  is a free variable):  $x = 4 - 2k$

**Parametric Solution:**  $(x, y, z) = (4 - 2k, k, 1)$  for any  $k \in \mathbb{R}$ .

### Example 3: Non-Homogeneous — INCONSISTENT (No Solution)

Solve the system:

$$x + 2y + z = 4$$

$$2x + 4y + 2z = 8$$

$$3x + 6y + 3z = 10$$

#### Step 1: Form Augmented Matrix $[A|B]$

$$\left| \begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 2 & 4 & 2 & 8 \\ 3 & 6 & 3 & 10 \end{array} \right|$$

#### Step 2: Reduce to Echelon Form

Apply  $R_2 \rightarrow R_2 - 2R_1$  and  $R_3 \rightarrow R_3 - 3R_1$ :

$$\left| \begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -2 \end{array} \right|$$

Swap  $R_2 \leftrightarrow R_3$ :

$$\left| \begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 0 & 0 & 0 & -2 \\ 0 & 0 & 0 & 0 \end{array} \right|$$

#### Step 3: Analyze Rank & Conclusion

- $\text{rank}(A) = 1$  (coefficient part has 1 non-zero row).
- $\text{rank}([A|B]) = 2$  (augmented matrix has 2 non-zero rows).

Row 2 yields the equation  $0x + 0y + 0z = -2$ , which means  $0 = -2$  (a physical contradiction).

**Conclusion:** Since  $\text{rank}(A) \neq \text{rank}([A|B])$ , the system is **INCONSISTENT** and has **NO SOLUTION**.

#### Example 4: Homogeneous System — NON-TRIVIAL Solutions

Solve the homogeneous system  $AX = 0$ :

$$x + 2y - 3z = 0$$

$$2x + 5y + 2z = 0$$

$$3x + 7y - z = 0$$

##### Step 1: Reduce Coefficient Matrix A

$$\left| \begin{array}{ccc} 1 & 2 & -3 \\ 2 & 5 & 2 \\ 3 & 7 & -1 \end{array} \right|$$

Apply  $R_2 \rightarrow R_2 - 2R_1$  and  $R_3 \rightarrow R_3 - 3R_1$ :

$$\left| \begin{array}{ccc} 1 & 2 & -3 \\ 0 & 1 & 8 \\ 0 & 1 & 8 \end{array} \right|$$

Apply  $R_3 \rightarrow R_3 - R_2$ :

$$\left| \begin{array}{ccc} 1 & 2 & -3 \\ 0 & 1 & 8 \\ 0 & 0 & 0 \end{array} \right|$$

##### Step 2: Check Rank & Solutions

$\text{rank}(A) = 2 < 3$ . Therefore, non-trivial solutions exist!

- From Row 2:  $y + 8z = 0 \Rightarrow y = -8z$
- Let  $z = k$  (where  $k \in \mathbb{R}$ ):  $y = -8k$
- From Row 1:  $x + 2(-8k) - 3k = 0 \Rightarrow x - 16k - 3k = 0 \Rightarrow x = 19k$

**Non-Trivial Solutions:**  $(x, y, z) = (19k, -8k, k)$  for  $k \in \mathbb{R}$ .

### Example 5: Homogeneous System — TRIVIAL Solution Only

Solve the system:

$$x + y + z = 0$$

$$x - y + 2z = 0$$

$$2x + y - z = 0$$

**Reduction to Echelon Form:**

Apply  $R_2 \rightarrow R_2 - R_1$  and  $R_3 \rightarrow R_3 - 2R_1$ :

$$\left| \begin{array}{ccc} 1 & 1 & 1 \\ 0 & -2 & 1 \\ 0 & -1 & -3 \end{array} \right|$$

Swap  $R_2 \leftrightarrow R_3$  and multiply new Row 2 by -1:

$$\left| \begin{array}{ccc} 1 & 1 & 1 \\ 0 & 1 & 3 \\ 0 & -2 & 1 \end{array} \right|$$

Apply  $R_3 \rightarrow R_3 + 2R_2$ :

$$\left| \begin{array}{ccc} 1 & 1 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 7 \end{array} \right|$$

**Conclusion:**

$\text{rank}(A) = 3 = n$ . Back substitution gives  $7z = 0 \Rightarrow z = 0$ ,  $y + 3(0) = 0 \Rightarrow y = 0$ , and  $x = 0$ .

**Unique Solution:** Only the **Trivial Solution**  $(x, y, z) = (0, 0, 0)$  exists.